

The Indolamine Serotonin 5-hydroxytryptamine (5-HT)

Learning objectives

Neurochemistry of 5-HT

- Synthesis
- Release
- Inactivation

5-HT Systems

- Anatomy (cell groups & pathways)
- Receptors
- Behavior

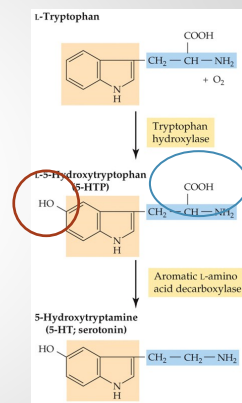
Serotonin is synthesized from the amino acid tryptophan

Tryptophan

- is obtained from the diet

Enzymatic synthesis

- Tryptophan hydroxylase (TPH2)
- Fe^{2+} , O_2 , BH_4
- AADC
- Pyridoxal phosphate (B_6)



PSYCHOPHARMACOLOGY 4e, Figure 6.1
© 2013 Oxford University Press

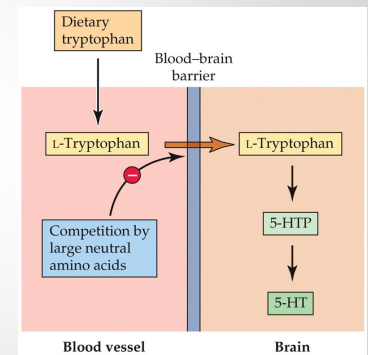
Tryptophan (TRP) and LNAAs compete for transport across BBB

High-protein, low-carbohydrate meal

- does not increase brain TRP levels or 5-HT synthesis

high-carbohydrate, low-protein meal

- Increases ratio of TRP : LNAAs
- enhances entry of TRP and stimulates 5-HT synthesis.



PSYCHOPHARMACOLOGY 4e, Figure 6.2
© 2013 Oxford University Press

TPH is a critical control point in 5-HT synthesis

TPH is the rate-limiting enzyme

TRP depletion

- TRP-free, high LNAA solution
- induces symptoms in people with major depression

p-chlorophenylalanine (PCPA)

- blocks 5-HT synthesis by irreversibly inhibiting TPH

5-HT is loaded into vesicles by VMAT2

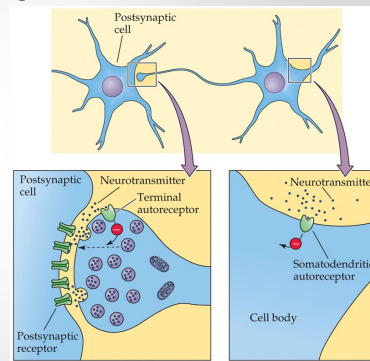
Vesicular monoamine transporter (VMAT)

- Different from SERT located on nerve terminal
- Blocked by reserpine
- Energy provided by a proton gradient
- ATP-driven H^+ pump acidifies vesicle lumen

Release of 5-HT is decreased by autoreceptor activation

5-HT_{1B/1D} terminal autoreceptors

5-HT_{1A} somatodendritic autoreceptors

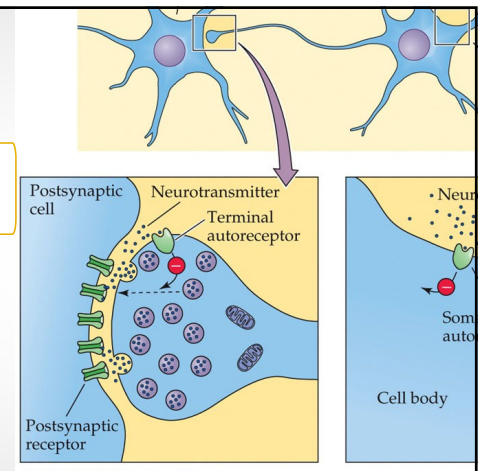


PSYCHOPHARMACOLOGY 4e, Figure 3.10
© 2013 Oxford University Press

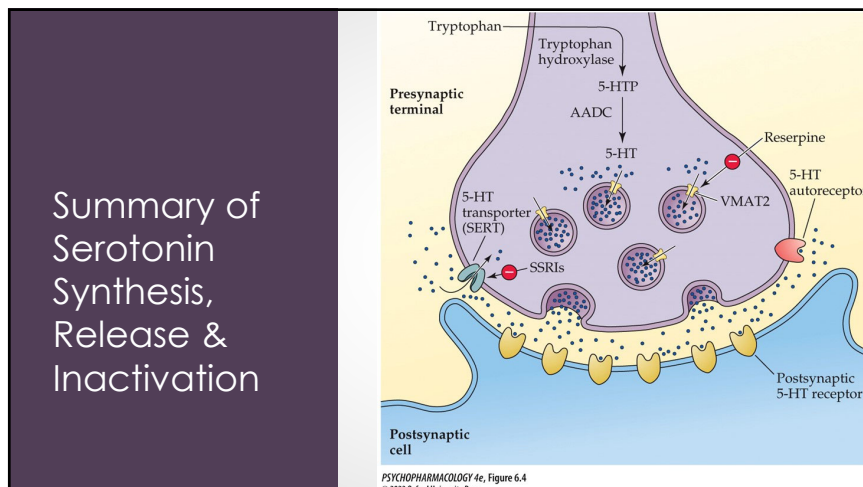
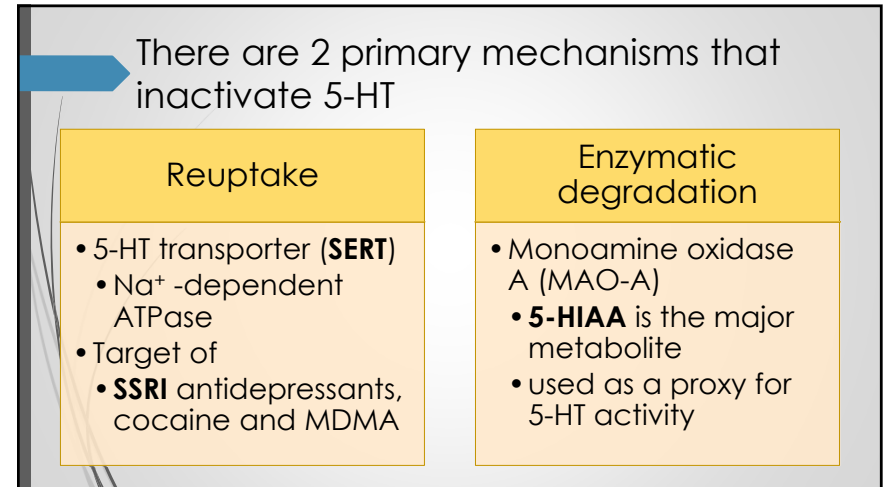
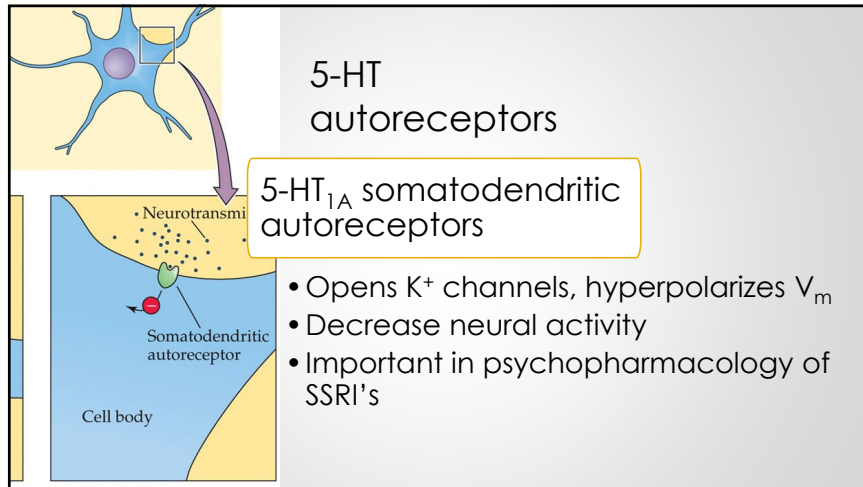
5-HT autoreceptors

5-HT_{1B/1D} terminal autoreceptors

- Inhibit 5-HT release
- decrease Ca^{2+} influx;
- decrease cAMP



PSYCHOPHARMACOLOGY 4e, Figure 3.10
© 2013 Oxford University Press



Pharmacological tools used to determine the role of 5-HT in behavior

Drug	Mechanism of Action
<i>para</i> -Chlorophenylalanine (PCPA)	General tryptophan hydroxylase inhibitor
Telotristat	Peripheral tryptophan hydroxylase inhibitor
Reserpine	Irreversible VMAT inhibitor
<i>para</i> -Chloroamphetamine and fenfluramine	5-HT releaser
MDMA	5-HT releaser
Fluoxetine and others	5-HT reuptake inhibitors (SSRIs)
Buspirone	5-HT _{1A} partial agonist
Ipsapirone	5-HT _{1A} partial agonist

Serotonin releasing agents (SRA's) act to increase 5-HT release

Para-chloroamphetamine (PCA, 4-CA)

- Neurotoxic, similar to 6-OHDA

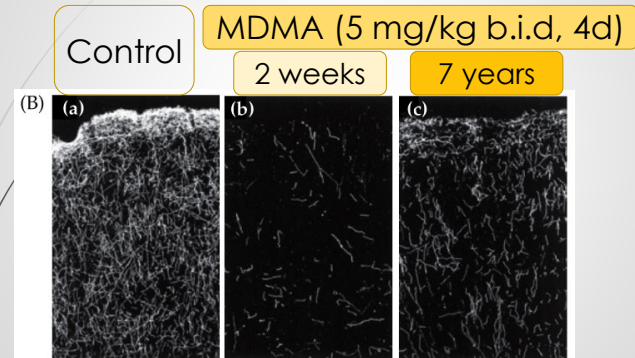
Fenfluramine

- in combo with phentermine (fen-phen)

MDMA, ecstasy, X, Molly

- Recreational drug, dangerous neurotoxin, or medication?

MDMA reduces 5-HT fiber density across widespread areas of cortex



Dance floor drug could treat PTSD



Researchers hope to test whether ecstasy can help treat soldiers and other sufferers of PTSD.

MDMA assisted psychotherapy for PTSD

MDMA

- 2 doses 3 – 5 weeks apart, together with weekly non-drug therapy

6 phase II clinical trials

- showed 38 of 72 (53%) subjects no longer meet criteria for PTSD diagnosis after 2 months

Durable remission

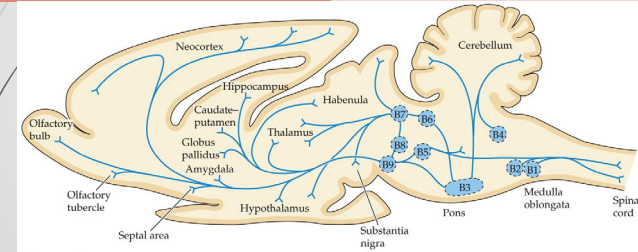
- mean follow-up 3.5 yr after MDMA
- in 14 of 19 (68%) of subjects (2 known relapse)

The Serotonin System Organization and Function

Almost all 5-HT neurons in the CNS are found along the midline of the midbrain

Raphé nuclei

- Dorsal raphé (DRN; B7)
- Medial raphé (MRN; B8)



There are at least 15 different types of 5-HT receptors, named 5-HT₁ – 5-HT₇

5-HT_{1A}

Inhibit adenylyl cyclase activity
Decreases cAMP

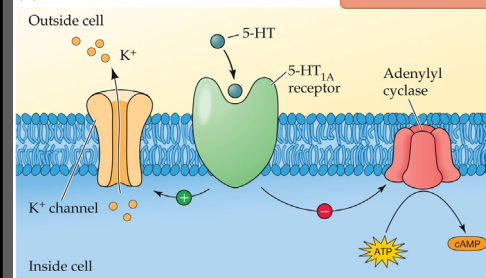
Increase g_K
Membrane hyperpolarization

5-HT_{2A}

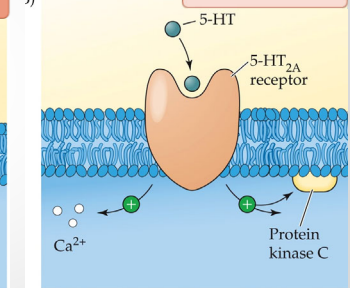
Coupled to PLC
Increase $[Ca^{2+}]_i$
Activates PKC

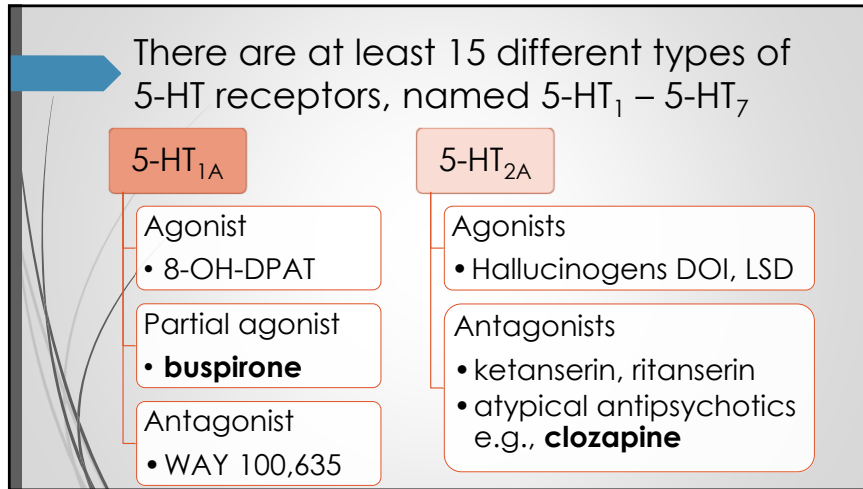
There are at least 15 different types of 5-HT receptors, named 5-HT₁ – 5-HT₇

5-HT_{1A}



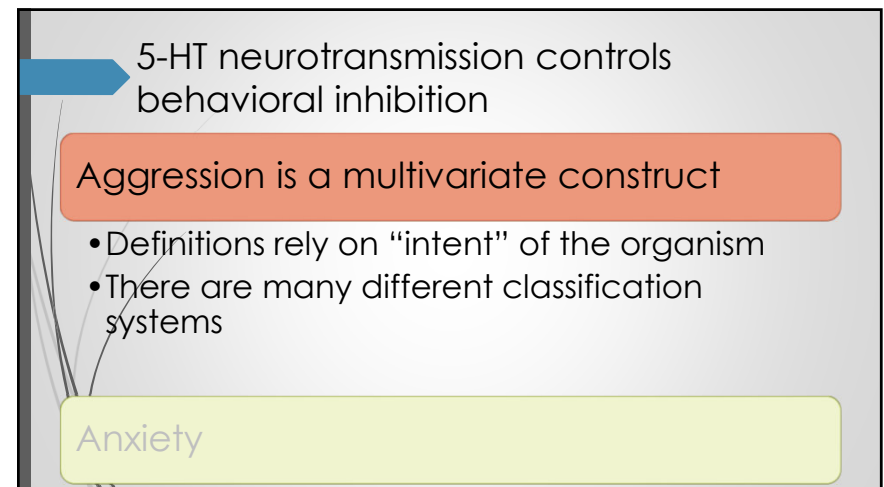
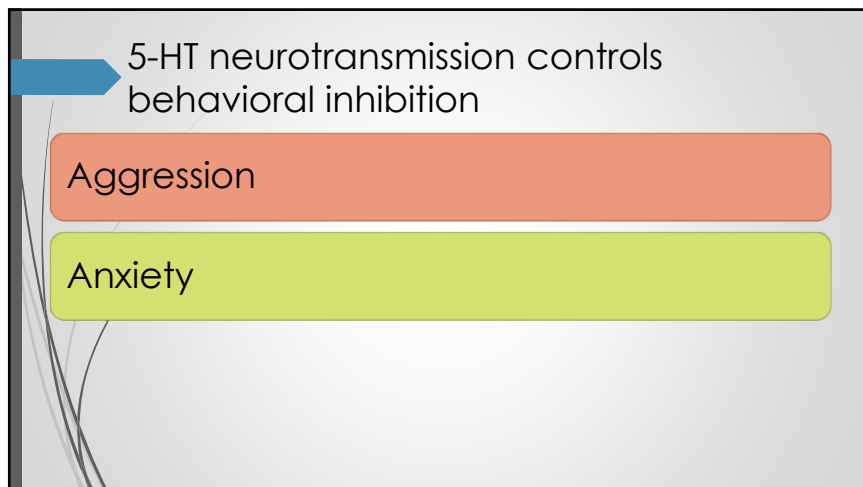
5-HT_{2A}





Pharmacological tools used to determine the role of 5-HT in behavior

Drug	Mechanism of Action
Buspirone	5-HT _{1A} partial agonist
Ipsapirone	5-HT _{1A} partial agonist
8-OH-DPAT	5-HT _{1A} full agonist
WAY-100635	5-HT _{1A} antagonist
DOI	5-HT _{2A} agonist
LSD	5-HT _{2A} agonist
Ketanserin and ritanserin	5-HT _{2A} antagonists
Triptans	5-HT _{1B/1D} agonists
Lasmiditan	5-HT _{1F} agonist
Ondansetron, granisetron, and palonosetron	5-HT ₃ antagonists



There are many different aggression classification systems

Human aggression can be classified as

Premeditated

- Instrumental
- Planned in advance

Impulsive

- Reactive
- Provoked
- “Hair trigger”

Animal aggression is inferred

Animal aggression is inferred

Types of Aggression

Category of aggression	Function and/or experimental conditions
Predatory	Attack of a predator on a prey animal
Antipredatory	Attack of a prey animal against its predator
Sex-related	Aggression that occurs in the context of a sexual interaction between two animals
Dominance	Aggression (usually between males) involved in the establishment or maintenance of rank within their group's dominance hierarchy
Maternal	Attack by a mother in the defense of her offspring
Territorial	Aggression that occurs in the context of <u>defending one's</u> territory, commonly modeled in the laboratory by using the <u>resident-intruder</u> test in which a strange male is introduced into the home cage of a resident male
Defensive	Attack elicited by a fear-inducing stimulus
Irritable	Attack elicited by extreme frustration or stress, sometimes modeled in the laboratory by the application of a brief electric shock

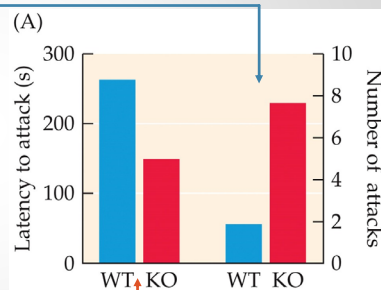
Source: After Nelson and Chiavegatto, 2001.

PSYCHOPHARMACOLOGY 3e, Box 6.2 Table
© 2019 Oxford University Press

Impaired 5-HTergic signaling increases behavioral aggression

Tph2 knock-out (KO) mice

- are virtually devoid of 5-HT in the brain
- are more **aggressive**
- are more **impulsive**



Alber M, Angoa-Pérez et al. 2012. *J Neurochem* 121: 974–984. © 2012 The Authors.
Journal of Neurochemistry
PSYCHOPHARMACOLOGY 4e, Figure 6.11 (Part 1)
© 2019 Oxford University Press

The 5-HT Deficiency Hypothesis:

Impaired 5-HTergic signaling increases behavioral aggression



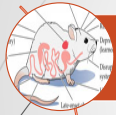
Animal studies of 5-HT



Human behavioral disorders

The 5-HT Deficiency Hypothesis:

Impaired 5-HTergic signaling increases behavioral aggression



Animal studies of 5-HT

- *Tph2* Knock-out mice
- 5,7-DHT lesions
- PCPA
- 5-HT_{1A} and 5-HT_{1B} KO mice

The 5-HT Deficiency Hypothesis:

Impaired 5-HTergic signaling increases behavioral aggression



Human behavioral disorders

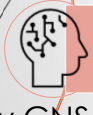
- Low 5-HIAA in people with anti-social or borderline personality disorders
- Alzheimer's disease
- Alcohol abuse
- Suicide

The 5-HT Deficiency Hypothesis:

Impaired 5-HTergic signaling increases behavioral aggression



Animal studies of 5-HT



Human behavioral disorders

Low CNS 5-HT is associated with more aggression

5-HT is also implicated in anxiety

5-HT_{1A} *partial agonists* are anxiolytic

- buspirone (BuSpar) & vilazodone (Viibryd)

5-HT_{1A} KO mice are more anxious

5-HT_{2A/2C} receptors are also involved

- The 5-HT_{2A/2C} agonist mCPP increases anxiety
- 5-HT_{2A} KO and 5-HT_{2C} KO reduces anxiety

5-HT neurotransmission controls
behavioral inhibition

Aggression

- Manipulations that lower 5-HT signaling generally impair behavioral control

Anxiety

- Without 1A, 2A or 2C receptors, behavioral control is impaired

Feeding

- 5-HT_{1B} and 5-HT_{2C} agonists (lorcaserin / Belviiq) are anorectics, induce hypophagia